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How autonomous research turns experimental experience into scientific understanding

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Autonomous science promises to turn experimental freedom into advances in scientific understanding[1,2]. How agents transform the evidence they choose to acquire into relationships that guide further research remains poorly understood. Here we investigate this transformation in programmable chemical worlds, connecting experimental choices, contemporaneous judgments and predictions under new interventions. Broad operating and predictive comparisons motivate repeated equilibrium and crystallization studies with 90 research sessions. These studies document prospective tests, evidence-based revisions and working relationships that guide further experiments. Following the same equilibrium study from a proposed test to a later forecast reveals a consequential transition: an experiment appropriately weakens a local explanation, then becomes a reason to extend a response plateau into conditions it did not resolve. In crystallization, three repeated studies establish usable procedures and explicitly recognize high purity, yet substantially underestimate purity in new conditions. Supported local discovery and unreliable transfer can therefore arise within the same research history. The findings locate a limit of autonomous scientific progress in how experimental support is carried from one judgment to the next. Experimental autonomy can support useful scientific relationships; extending those relationships requires preserving what the evidence establishes and which possibilities remain open.

The central promise of autonomous science is that greater experimental agency will expand scientific understanding. An agent that chooses operations, measurements and the next experiment also shapes the experience from which its judgments form: which phenomena become visible, which hypotheses encounter a test and which results become grounds for an explanation. Autonomous research thus changes both the experiments machines can perform and the process through which evidence becomes knowledge.

Robot scientists and self-driving laboratories have demonstrated hypothesis testing, materials discovery and reaction optimization[3–6]. Language-model agents extend the connection between scientific instructions, tools and experimental operations[7,8]. These advances sharpen a fundamental tension: real experimental feedback does not automatically become reliable scientific understanding. A successful experiment establishes an outcome under particular conditions; deciding which relationship it supports and where that relationship applies requires further judgment. An effective procedure and a correct prediction each reveal part of scientific achievement, but neither alone specifies what the research has established[9,10].

We use ChemWorld to examine how agents turn self-organized experimental experience into relationships used in subsequent research. Agents act on evolving virtual samples, while an evaluator independently controls compatible world laws and public information. Broad comparisons provide the initial puzzle: operating progress, local prediction and transfer to new conditions need not improve together. Repeated studies with contemporaneous notebooks then connect the formation, revision and later use of specific judgments. The central observation is that a study can establish a useful local relationship while changing or extending its evidential meaning in a subsequent judgment. Discovery and misunderstanding can emerge at different points in the same research history.

Research as a traceable experimental process

ChemWorld combines materials, operations and instruments into programmable chemical systems with persistent state. Researchers choose quantities, operation sequences, measurements, repetitions and subsequent experiments; operations can alter samples irreversibly, consume resources or be rejected before changing physical state. Evaluators retain the executed transitions and underlying responses. These are controlled virtual chemical worlds rather than claims about experimentally validated substances in a physical laboratory.

This arrangement extends the evidence available from knowledge tests and supplied-data discovery tasks[11,12]. Interactive discovery environments already connect action and observation[13–16]. Here, composable chemical operations make the same researcher's choice of evidence, evolving public account and use of acquired relationships jointly inspectable (Fig. 1).

The retained broad study comprises 240 campaigns across six system families and 60 additional equilibrium campaigns from four other model configurations. It supplies the operating and predictive context. In electrochemistry, optimization delivers the higher-scoring recommendation in 26 of 30 matched comparisons, but its predictions are better in 13 and worse in 13 of those comparisons. Research resources and prior information also have positive effects in particular settings: increasing the partitioning budget reduces organic-phase-fraction prediction error from 0.08291 to 0.02379, and aligned reaction priors improve prediction in four of five worlds under each

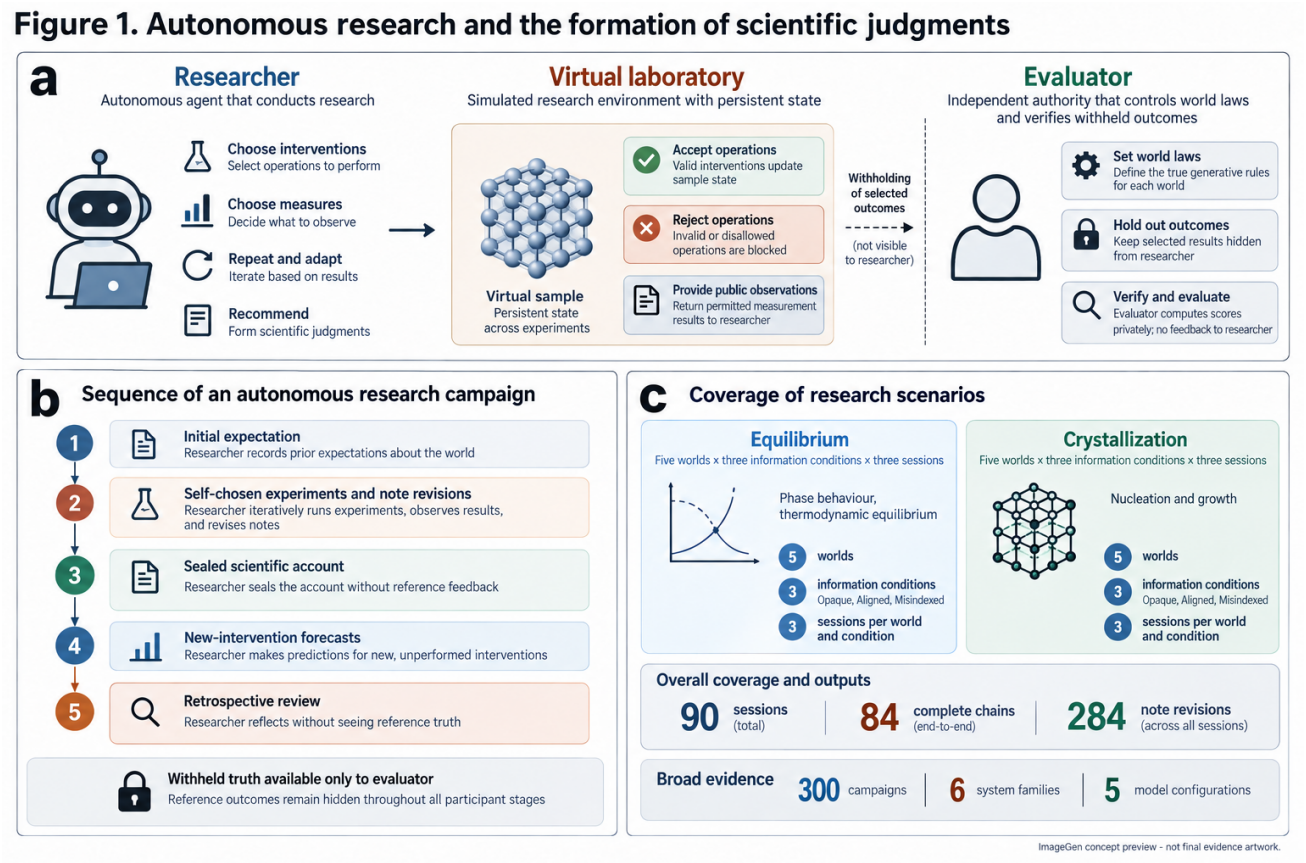


Figure 1 | IMAGEGEN TEMPORARY PREVIEW. Generated marks and geometry are illustrative; retained tables and text carry the numerical evidence. Connecting autonomous research to the formation and use of judgments. a, Researchers choose experiments on persistent virtual samples; evaluator records separately identify committed operations, observed responses and world reference outcomes. b, Contemporaneous notes precede the sealed scientific account, forecasts and retrospective review. Only the evaluator receives withheld reference outcomes. c, The broad study and the repeated process cohort are separate evidence layers. The process design has three sessions per world and information condition; totals retain the six incomplete chains. The five worlds recur across information conditions and do not constitute new world samples in each arm.

of two research goals. These are comparisons between independent campaigns, not within-researcher learning curves.

To examine judgment formation during experimentation, a separate cohort covers equilibrium and crystallization, each in five worlds, three information conditions and three fresh sessions per condition. Agents receive no extra instance information, an aligned partial prior, or a specified mismatched prior. All sessions use the same Sol-medium model configuration and a twelve-batch research allowance. The cohort contains 90 terminal sessions, 84 complete research-and-assessment chains and 284 notebook revisions. Failures remain in the planned denominator. After experimentation, the agent seals its scientific account, predicts fixed new interventions without further laboratory access, and then reflects without seeing prediction truth. World parameters and observation mappings are fixed across repeats; model sampling randomness is not controlled.

A local test acquires a broader role

One equilibrium study provides a continuous account of how an experiment's scientific role changes. A nearly flat reagent-loading response leads the researcher to propose that another accessible component, rather than the nominal reagent, might be the active coordinate. Before the final two experiments, it predicts that this alternative should produce a substantial response. The researcher then measures a component-only sample and a large component addition to a small reagent charge. Both remain near the observed plateau. Its next note withdraws support for the tested mapping, retains competing explanations and limits its account to the measured conditions (Fig. 2a). The experiment has resolved part of an explicit scientific question without identifying a unique hidden mechanism.

We follow this same study into its sealed predictions (Fig. 3a). Asked about an extremely dilute condition, the researcher explicitly cites the component-only result as a reason to shrink its forecast towards the observed plateau. It predicts a dissociation fraction of 0.092, with an 80% interval of 0.043–0.155; the withheld reference mean is 0.51730. The earlier experiment opposed a particular component-mapping explanation, but it did not determine whether the plateau would persist at extreme dilution. The result is first used to make an appropriately bounded revision, then used to favour a continuation that the completed tests had not distinguished.

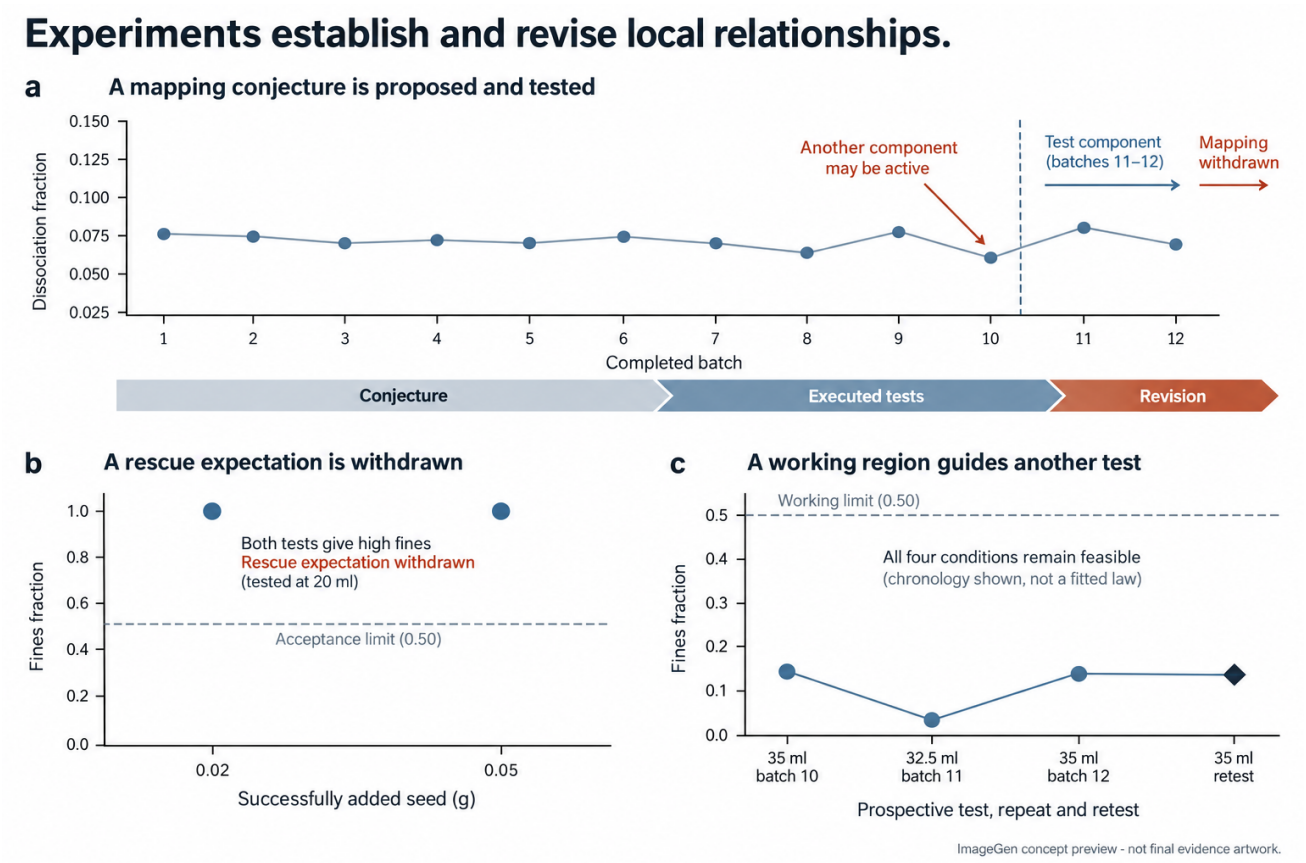


Figure 2 | IMAGEGEN TEMPORARY PREVIEW. Generated marks and geometry are illustrative; retained tables and text carry the numerical evidence. Prospective tests and bounded working relationships. a, An equilibrium agent records a component-mapping conjecture before the final two batches, executes both perturbations and subsequently narrows its account. Points show public final-assay dissociation fractions; text condenses the corresponding public notes. b, A crystallization seed-rescue conjecture is followed by two tested seed doses and a recorded withdrawal. The horizontal line is the task's fines limit. c, A proposed 32.5 ml test is followed by a 35 ml repeat and a separate recommendation retest. Points show public outcomes except for the explicitly labelled noiseless retest. The connected points indicate experimental order, not a fitted response law. Cases are retrospectively selected; complete session outcomes are retained in the supplementary tables. The equilibrium test in panel a continues into the forecast in Fig. 3a: both panels follow the same session. Recovery values retain the implemented initial-charge definition described in Methods.

The scientific account still lists buffering, saturation and unidentified coupling as competing explanations. During retrospective review, without receiving the reference answer, the researcher acknowledges that its forecast interval underrepresents these unresolved possibilities. The departure therefore occurs despite an explicit recognition of uncertainty. The observable change lies in the weight and scope assigned to the earlier evidence when constructing a new judgment. The notebook, committed operations and stated forecast rationale locate that transition within one research history; they do not identify a private belief state or the independent causal effect of a particular statement.

Working relationships support further inquiry

A crystallization study provides a more direct prospective test. After a concentrated recipe fails the fines requirement, the researcher proposes increasing seed mass to rescue particle quality. The two relevant batches share the principal preparation and cooling operations, with successful seed additions of 0.02 and 0.05 g. Both return a fines fraction of 1.0. The next note explicitly withdraws the rescue hypothesis for the tested condition (Fig. 2b). The agent ultimately recommends another recipe that passes independent retesting. A separate study withdraws an expected monotonic loading effect after a lower charge worsens fines. Scientific revision is therefore observable even when the proposed route to better operation fails.

Experimental relationships can also become working guides for further inquiry. In another crystallization session, successes and failures motivate a volume-window search. A note proposes a test at 32.5 ml after obtaining a feasible process at 35 ml. The new test retains the other principal operations and remains feasible; the following experiment repeats 35 ml, and the recommendation is independently retested (Fig. 2c). This sequence establishes usable recipes and motivates a subsequent experiment. Earlier failures at lower volumes also change seed or thermal history, so they do not independently identify a concentration threshold.

These records also contain explicit distinctions between an observed recipe trend and an independently identified temperature coefficient, a supported region and a universal solvent ranking, and unresolved small differences and a repeatable improvement. Such limits are part of the relationship

being established. The structured process records in Supplementary Information connect each focal expectation to its executed test, subsequent revision, supported scope and later use, alongside record availability and outcomes for all ninety sessions. This places the selected trajectories within the study without turning them into an estimated frequency of a single mechanism.

Repeated studies locate the limits of transfer

The repeated equilibrium cohort places the focal study within a broader distribution of outcomes. Aligned priors reduce five-world mean error over the other nine queries from 0.00907 to 0.00512, but increase error over the three most dilute queries from 0.09880 to 0.13205 (Fig. 3b). Local improvement occurs in all five worlds and dilute deterioration in four. Empirical coverage of nominal 80% dilute intervals falls from 73.85% to 19.48%. Mismatched-prior errors are 0.00531 and 0.14848, with dilute coverage of 14.89%. These results preserve a regional reversal of prior benefit; they do not reproduce the original study's approximately 8.7-fold magnitude. The current aligned-to-opaque ratio is about 1.34. The later collection of all mismatched sources is retained as a cohort difference in Methods.

All 45 equilibrium sessions remain above 1 mM in their positively loaded source assays. Nevertheless, similar plateau observations support different forecast constructions. In repeated studies of the same world, opaque sessions produce either a near-plateau estimate or a much larger, accurate dissociation forecast, whereas the aligned sessions consistently remain closer to the plateau (Fig. 3c). Another opaque session introduces an effective soluble-concentration cap when answering the prediction questions, obtains an accurate dilute forecast and subsequently acknowledges that the mechanism was only one candidate in its earlier account.

Direct sampling is therefore not a requirement for legitimate or successful extrapolation. In the separate five-model study, Sol/Opaque extrapolates accurately without reaching the dilute region, while an Astra campaign first observes a large dissociation increase at low concentration and then predicts a new dilute condition accurately. These paths preserve the distinction between a correct forecast, evidence acquired during research and experimental discrimination among explanations. Prediction scores alone collapse these differences.

Recognized relationships can change in transfer

Crystallization exposes a complementary issue: a researcher may correctly recognize a stable response but substantially alter it when predicting new conditions. In three sessions under the same world and mismatched information, each agent finds a recipe that passes recommendation retesting. Each also identifies high crystal purity in its scientific account. Mean source purities are 99.09%, 98.18% and 99.09%, yet mean forecasts for the new recipes are 86.92%, 86.33% and 89.92%; their common reference mean is 98.45% (Fig. 4a).

The first session offers a particularly clear within-research contrast. Its source account describes substantial impurity rejection and a bounded feasible recipe region. When forecasting, it introduces adjustments for solvent, impurity occlusion, concentration and thermal history. Purity mean absolute error is 0.11529, and none of its twelve purity intervals covers the reference. New recipes legitimately require judgment about changed conditions. Here, however, those adjustments move predictions away from a response that the experiments and the researcher's own account had established locally and that remains high in the new conditions.

Across all 45 crystallization sessions, recommendations satisfy the particle, purity and fines requirements in 10 of 15 opaque, 9 of 15 aligned and 14 of 15 mismatched-prior sessions. Their five-world mean purity errors are 0.01537, 0.04120 and 0.05964, respectively. The retained joint quality-and-recovery counts are 9, 7 and 14 of 15 (Fig. 4b,c). A metric-definition check identifies a mismatch: the public recovery description uses target product before separation, whereas the observed and scored full-process response uses initial reagent charge. Recovery forecast scores and the joint operational readout are therefore retained as protocol-affected diagnostics rather than clean measures of scientific transfer. Purity and fines do not use that denominator. The mismatched information also leaves the frequently used solvent S2 unchanged and belongs to the later cohort; its outcome cannot establish a benefit of incorrect knowledge.

The earlier crystallization study found 335 of 360 purity forecasts below reference despite source and reference purity near 98.5%. The contemporaneous records add a more specific result: the high-purity relationship was explicitly recognized by the same researchers who subsequently altered it in prediction. Together with the equilibrium trajectory, this shows two distinct transitions. A local test can be assigned a broader scope than it resolved; a locally established response can receive less weight when new conditions are considered. Neither transition is equivalent to failing to obtain useful evidence in the first place.

The evidential status of a judgment can itself be mistaken

Some departures occur before transfer. One agent proposes testing about 2 g of cumulative seed, interpreting the dose limit as a per-operation rather than a per-batch constraint. It requests forty additions of 0.05 g, but only the first succeeds; the other thirty-nine are rejected. A later note nonetheless states that raising seed from 0.05 to about 2 g did not reduce fines and excludes seed mass as the limiting control. The final scientific account and retrospective review retain that conclusion (Extended Data Fig. 1).

The claimed dose comparison was never completed. Its failure cannot provide negative experimental evidence about a large dose. This trajectory locates the conversion of an intended test into an apparently established exclusion. Another agent correctly records repeated seed-addition rejection and limits subsequent dosing accordingly. Execution failures therefore do not inevitably become scientific misstatements. The first case concerns interpretation of execu-

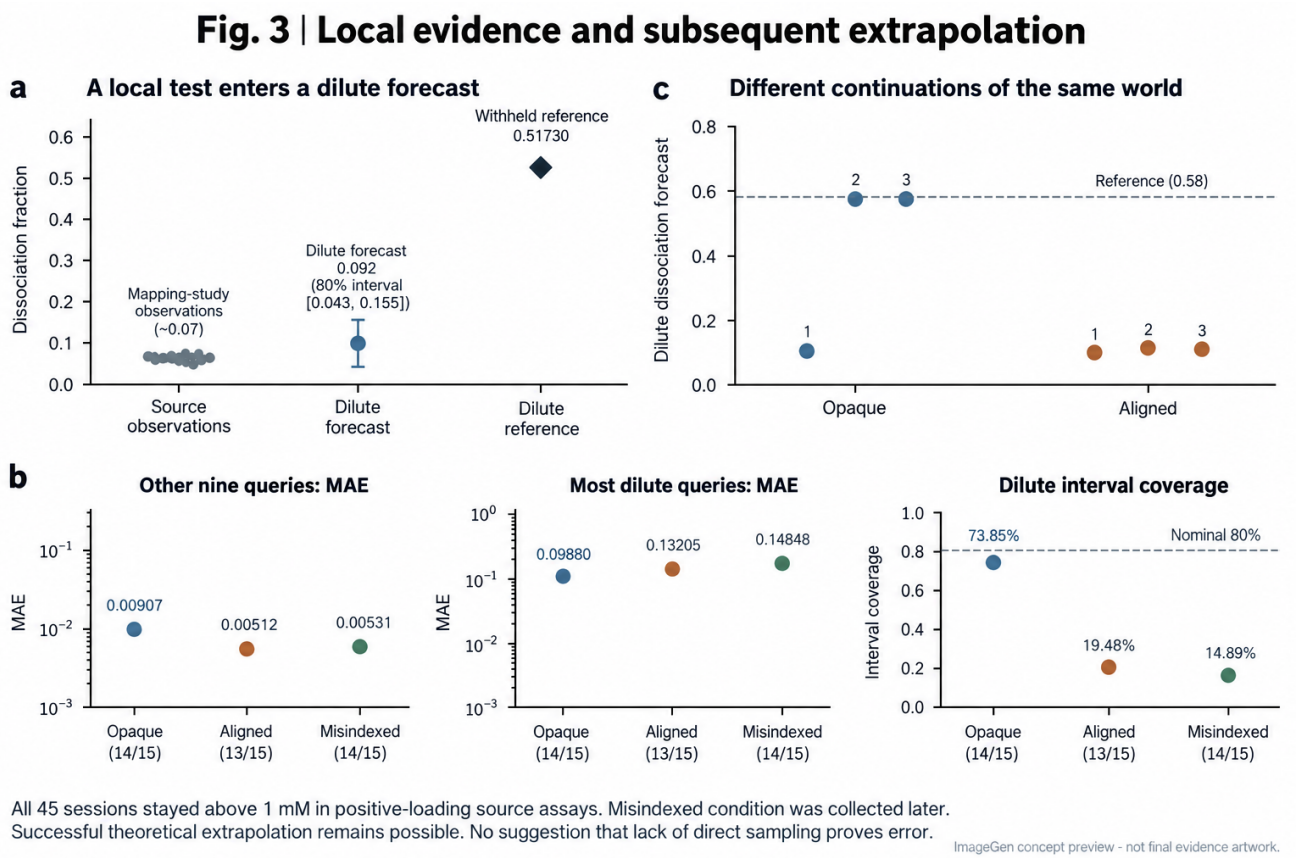


Figure 3 | IMAGEGEN TEMPORARY PREVIEW. Generated marks and geometry are illustrative; retained tables and text carry the numerical evidence. Local evidence and subsequent extrapolation. a, Public dissociation observations in the component-mapping study, followed by the sealed dilute-query forecast, its original 80% interval and the five-observation reference mean. b, Mean absolute errors for the three most dilute queries and the other nine, and empirical interval coverage for the dilute group. The preview shows five-world-equal summaries; missing forecasts are not assigned zero. Individual session values are retained in Supplementary Information. Opaque, Aligned and MisIndexed have 14, 13 and 14 valid forecasts out of 15 planned sessions. The mismatched arm belongs to the later cohort. c, Forecasts from three fresh sessions per information condition in one fixed world, with the shared reference mean. Repeats are model sessions, not independent world or measurement-noise samples. Panel a follows the same research trajectory as Fig. 2a; the local mapping test and dilute forecast are successive judgments within one study.

tion state; it does not by itself explain other forms of generalization error or prove that the mistaken note caused subsequent actions.

Similarly, recognizing uncertainty need not settle an operating choice. One researcher states a preference for fines margin but ultimately selects the highest observed feasible recovery near the boundary, which fails noiseless retesting. Another sacrifices some observed recovery for margin and passes (Extended Data Fig. 2). The task rewards observed feasible recovery, so a boundary choice is not automatically irrational. Its relation to the researcher's own stated decision rule remains informative.

Experimental autonomy and the growth of understanding

Autonomous research can form useful scientific relationships while leaving their later scope and use unsettled. The observed agents formulate prospective tests, revise expectations after contrary results and develop working relationships that organize further experiments. Following those achievements into new judgments reveals that supported local discovery and unreliable transfer can occur within the same study. Experimental progress can change what a researcher knows about tested conditions without equally resolving what it should conclude about another intervention.

This result brings the formation and continued use of scientific understanding into the study of autonomous science. Experimental design and strong inference describe how tests can discriminate among explanations[17–19]. The present trajectories additionally reveal how an executed test acquires an explanatory role and how unresolved alternatives enter a subsequent quantitative judgment[20,21]. In the equilibrium case, a legitimate revision and an unsupported extension draw on the same experimental result. In crystallization, an explicitly recognized relationship is changed during transfer. The rejected-dose case locates a different departure, where an intended experiment is represented as completed evidence. Their unity lies in the conversion of research experience into grounds for further judgment.

The process cohort uses one model configuration, fixed worlds and a prompted notebook workflow. Interpretation is retrospective and by one analyst; the selected transitions establish observable cases, not their population frequency

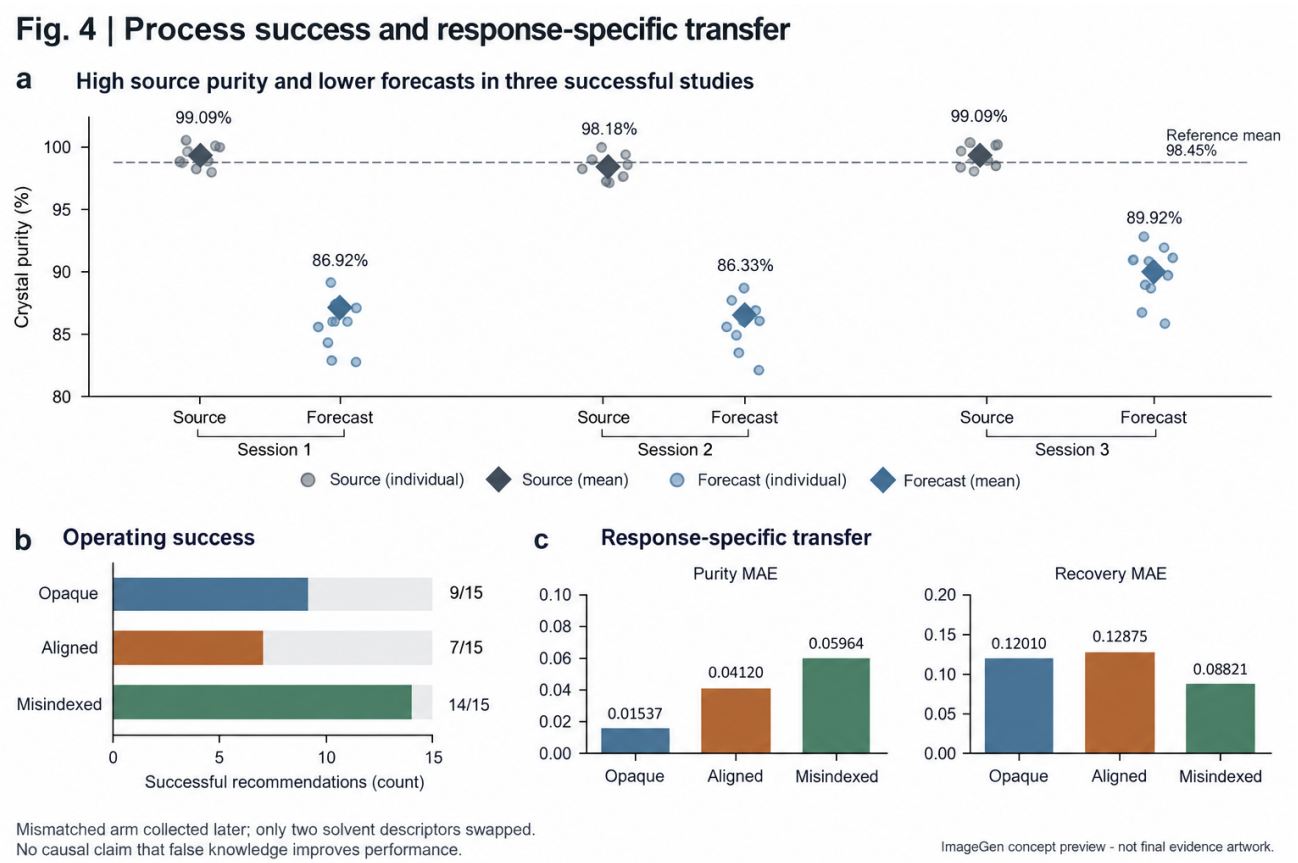


Figure 4 | IMAGEGEN TEMPORARY PREVIEW. Generated marks and geometry are illustrative; retained tables and text carry the numerical evidence. Process success and response-specific transfer. a, Three crystallization sessions in one fixed world and information condition. Small points are individual source purity observations and withheld recipe forecasts; larger markers indicate their means. Horizontal lines show the reference mean. All three recommendations pass the original retest criteria. b, Recommendation success counts for all 45 crystallization sessions. c, Five-world-equal purity and recovery prediction errors by information condition; individual session values are retained in Supplementary Information. Outcomes from the earlier and extension cohorts retain their separate provenance. Success requires particles, purity of at least 0.80, fines of at most 0.50 and seed-excluded recovery of at least 0.10. Recovery MAE and the joint counts in panels b,c are retained protocol-affected diagnostics: the prompt describes stage recovery, while observation and scoring use initial reagent charge. They are not clean measures of recovery understanding. Particle/purity/fines-only counts are 10/15, 9/15 and 14/15.

or a common causal mechanism. Missing intermediate statements are not evidence that a judgment never occurred. The recovery-definition mismatch further limits the crystallization protocol, although it does not directly change purity or fines. A matched quench comparison establishes physical consequences of an operation in fixed recipes (Extended Data Fig. 3); no agent receives the intervention results or revises its research, so it cannot explain the formation of an agent's account. Independent case review and targeted comparisons remain necessary for stronger causal or cross-model claims.

The scientific implication is that expanding experimental agency and expanding reliable understanding are connected but distinct achievements. Evidence can revise a local explanation, support a useful procedure and still enter a later judgment with a different scope or weight. Scientific progress therefore depends on how a researcher carries experimental support across successive questions. ChemWorld makes this continuity observable: what an agent discovers is inseparable from the research through which a relationship gains support and the further conditions under which it is used.

References

1. G. Tom et al. Self-Driving Laboratories for Chemistry and Materials Science. *Chemical Reviews* **124** 9633–9732 (2024). https://doi.org/10.1021/acs.chemrev.4c00055.
2. D. A. Boiko, R. MacKnight, B. Kline, and G. Gomes. Autonomous chemical research with large language models. *Nature* **624** 570–578 (2023). https://doi.org/10.1038/s41586-023-06792-0.
3. R. D. King et al. The Automation of Science. *Science* **324** 85–89 (2009). https://doi.org/10.1126/science.1165620.
4. B. Burger et al. A mobile robotic chemist. *Nature* **583** 237–241 (2020). https://doi.org/10.1038/s41586-020-2442-2.
5. B. P. MacLeod et al. Self-driving laboratory for accelerated discovery of thin-film materials. *Science Advances* **6** eaaz8867 (2020). https://doi.org/10.1126/sciadv.aaz8867.
6. N. J. Szymanski et al. An autonomous laboratory for the accelerated synthesis of inorganic materials. *Nature* **624** 86–91 (2023). https://doi.org/10.1038/s41586-023-06734-w.
7. A. M. Bran et al. Augmenting large language models with chemistry tools. *Nature Machine Intelligence* **6** 525–535 (2024). https://doi.org/10.1038/s42256-024-00832-8.
8. T. Dai et al. Autonomous mobile robots for exploratory synthetic chemistry. *Nature* **635** 890–897 (2024). https://doi.org/10.1038/s41586-024-08173-7.
9. A. N. Elmachtoub and P. Grigas. Smart Predict, then Optimize. *Management Science* **68** 9–26 (2022). https://doi.org/10.1287/mnsc.2020.3922.
10. G. Shmueli. To Explain or to Predict? *Statistical Science* **25** 289–310 (2010). https://doi.org/10.1214/10-sts330.
11. A. Mirza et al. A framework for evaluating the chemical knowledge and reasoning abilities of large language models against the expertise of chemists. *Nature Chemistry* **17** 1027–1034 (2025). https://doi.org/10.1038/s41557-025-01815-x.
12. B. P. Majumder et al. DiscoveryBench: Towards Data-Driven Discovery with Large Language Models. *International Conference on Learning Representations* **2025** 4556–4579 (2025). https://proceedings.iclr.cc/paper_files/paper/2025/file/0d70af566e69f1dfb687791ecf955e28-Paper-Conference.pdf.
13. R. Wang, P. Jansen, M.-A. Côté, and P. Ammanabrolu. ScienceWorld: Is your Agent Smarter than a 5th Grader? *Proceedings of the 2022 conference on empirical methods in natural language processing* 11279–11298 (2022). https://doi.org/10.18653/v1/2022.emnlp-main.775.
14. P. Jansen et al. DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. *Advances in neural information processing systems* **37** (2024). https://doi.org/10.52202/079017-0324.
15. K. Gandhi et al. BoxingGym: Benchmarking progress in automated experimental design and model discovery. (2025). https://arxiv.org/abs/2501.01540.
16. Y. Chen, P. Piękos, M. Ostaszewski, F. Laakom, and J. Schmidhuber. PhysGym: Benchmarking LLMs in Interactive Physics Discovery with Controlled Priors. *Advances in Neural Information Processing Systems* **38** (2025). https://doi.org/10.52202/085713-1556.
17. D. V. Lindley. On a Measure of the Information Provided by an Experiment. *The Annals of Mathematical Statistics* **27** 986–1005 (1956). https://doi.org/10.1214/aoms/1177728069.
18. K. Chaloner and I. Verdinelli. Bayesian Experimental Design: A Review. *Statistical Science* **10** 273–304 (1995). https://doi.org/10.1214/ss/1177009939.
19. J. R. Platt. Strong Inference. *Science* **146** 347–353 (1964). https://doi.org/10.1126/science.146.3642.347.
20. T. Gneiting and A. E. Raftery. Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association* **102** 359–378 (2007). https://doi.org/10.1198/016214506000001437.
21. Y. Ovadia et al. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. *Advances in Neural Information Processing Systems* **32** (2019). https://proceedings.neurips.cc/paper_files/paper/2019/file/8558cb408c1d76621371888657d2eb1d-Paper.pdf.

Methods

Experimental worlds and public interfaces

ChemWorld compiles compatible process components into stateful chemical environments. A public contract defines available operations, their legal domains, instruments, resource accounting and task objectives. Private parameters determine the instantiated response. Researchers can access the laboratory interface and its public records but not evaluator state, simulator code, private parameters or withheld outcomes. Samples retain the consequences of previous operations. Measurements may consume material. Candidate transitions are validated before commitment; rejected operations do not become physical interventions, although attempts can consume contract-defined allowances. Exact replay verifies recorded environment transitions, not regeneration of the language model's decisions.

The chemical families in the broad study are electrochemistry, phase partitioning, reaction and thermal processing, aqueous equilibrium, reaction-coupled crystallization and purification. They are virtual systems with deliberately controlled scope. Their anonymous material identities and qualitative chemical mechanisms do not establish predictive validity for specific real substances. Equilibrium response channels and crystallization models are described in the retained protocol evidence and supplementary tables. The process study preserves the relevant earlier world laws, information contracts and query definitions while generating references on its own frozen execution surface.

Broad study and evidence layers

The original cross-system study contains 240 Sol-medium campaigns: 60 electrochemical, 30 partitioning, 60 reaction, 45 equilibrium and 45 crystallization or purification campaigns. The equilibrium campaigns span three prior loci with fifteen campaigns each; crystallization contributes thirty and purification fifteen. Electrochemistry, partitioning and crystallization include twelve- and twenty-four-batch budgets. Reaction and electrochemistry include discovery and optimization commissions. Three information conditions recur across five instantiated worlds in each relevant comparison.

Under the original completion rules, 238 of these 240 campaigns are recorded as protocol-conforming; 3,597 of 3,600 planned final assays and all 720 scientific-account, forecast and retrospective stages are retained. One crystallization campaign exhausts solvent after eleven final assays; one purification campaign starts twelve batches but terminates only ten for final assessment. Their valid forecasts remain included. An additional 60 equilibrium campaigns comprise fifteen each from Luna, Terra, GPT-5.5 and Astra, requested at medium reasoning effort, with 720 source batches and 240 assessment stages. The combined broad pool is therefore 300 campaigns, not 300 plus a further 75: the five-model equilibrium comparison includes fifteen Sol campaigns already counted in the original 240. Public model labels denote requested configurations and do not imply equal realized computation.

Earlier notebook feasibility and selected-case studies informed process measurement. The six-case study and eighteen fresh repeats contain 24 sessions and are not pooled into either the broad study or the present 90-session cohort. The retained studies include development and exploratory evidence. Manuscript integration does not change their original qualification status. Complete block counts, earlier recoveries and all process-cohort dispositions are provided in Supplementary Information.

Repeated process cohort

The cohort contains two prospectively specified blocks. The first has thirty equilibrium sessions, covering five worlds and Opaque/Aligned conditions with three repetitions, and eighteen crystallization sessions covering the first, middle and last original world indices under the same two conditions and repetition count. Crystallization world selection is fixed before execution, rather than based on observed error. The extension adds fifteen MisIndexed equilibrium sessions, fifteen MisIndexed crystallization sessions, and twelve Opaque/Aligned crystallization sessions in the other two worlds. The descriptive combination covers two systems, five worlds, three information conditions and three sessions, for ninety planned sources.

Each source uses a fresh workspace and model context, the Sol-medium configuration and a twelve-batch allowance. The model chooses recipes, measurements, repetitions and operation grouping. World parameters, initial states and keyed observation mappings are fixed across repetitions. Model sampling seeds are unavailable; these are repeated model sessions in fixed worlds, not a controlled model-seed study. Identical recipes can encounter identical keyed measurements, so repeated encounters must not be interpreted as independent noise realizations.

Four isolated workers consume a predetermined queue. The first block interleaves systems and varies world/arm order across rounds; the extension rotates world and arm order before removing already covered cells. Completion order need not match launch order. Each source launches once. No failed source is replaced and no favorable outcome determines inclusion. Provider or participant failures remain terminal dispositions; an execution, reference or replay defect would stop new launches. Source deadlines are 5,400 s for equilibrium and 3,600 s for crystallization. Resource limits remain those of the fixed original engines. In particular, equilibrium permits 180 operation attempts, twelve intermediate instruments,

twelve final assays, 0.24 mol reagent and 0.72 L solvent. Crystallization uses its twelve-batch resource envelope with a 720-attempt ceiling and separately bounded materials and instruments.

The first block completes all 576 source batches and 162 of 174 assessment stages, with 44 complete chains and 145 notebook versions. The extension completes 503 of 504 batches and 136 of 141 stages, with 40 complete chains and 139 versions. Combined counts are 1,079 of 1,080 batches, 298 of 315 stages and 84 complete chains. There are 86 valid forecast stages: 41 in equilibrium and 45 in crystallization. The six incomplete chains remain in the source denominator; missing forecasts are neither errors of zero nor replaced by later attempts. All ninety recorded source trajectories and all forty-five crystallization recommendation retests pass exact replay.

Information conditions and contemporaneous records

Opaque provides the operational interface, task and public context without the additional instance dossier. Aligned provides the fixed partial information appropriate to the block. In equilibrium, this is an effective-acidity interval and a local dilution relationship at a specified anchor. MisIndexed presents another world's archival relationship in the same format, with the same qualification and precision. In crystallization, Aligned gives anonymous nominal material descriptors rather than complete instance laws. MisIndexed exchanges the descriptors of solvents S1 and S3, leaving S0, S2 and catalyst descriptors unchanged. It therefore does not corrupt every property the researcher may use.

The two blocks are sequential cohorts. All mismatched-prior sources belong to the extension, while Opaque/Aligned crystallization sources span both periods. Information-arm differences involving those sources can include time or provider drift. They are reported descriptively, not as a concurrently randomized estimate of the effect of mismatching information.

The same notebook role and version tools are available in every source. The role requests an initial concise testable expectation, where one exists, before physical experimentation. Subsequent updates are chosen by the model, without a per-batch write requirement. Notes are not automatically injected and no hints about dilution, quench, prior outcomes or the control study are supplied. The recording role can influence research and is not claimed to be passive. All ninety initial notes precede physical actions. The 284 retained revisions include their public-event cursor and time, allowing a statement to be located relative to available observations. There are no explicit notebook-read tool calls; this does not demonstrate that note content was inaccessible or unused within the session.

Sealed assessments and reference outcomes

After research and any required operating recommendation are sealed, the agent gives an unconstrained scientific account, then forecasts twelve fixed new-batch recipes, then answers retrospective questions. Equilibrium adds a structured supplemental stage. The notation K1, Q, K2 and EQS in the evidence records identifies these stages, respectively. Prediction queries are not exposed during source research. The forecast stage permits no new laboratory operations, and reference truth is withheld throughout all participant stages. A proposal first made in retrospective review is not counted as an executed experiment or backdated to source research.

For the process cohort, 300 equilibrium reference batches cover five worlds, twelve queries and five fixed observation repeats. Sixty crystallization reference batches cover five worlds and twelve queries. The extension reuses the first block's 336 references and adds only the twenty-four missing crystallization batches. References reside outside participant workspaces; their earlier generation in this cohort does not provide participant feedback. Each crystallization recommendation is separately retested from its committed recipe. Requested but rejected actions do not become successful material additions in that recipe.

Equilibrium point errors compare forecasts with the mean of five reference observations, and interval coverage checks each of those observations. Crystallization errors and intervals use the noiseless full-process observation response before final-assay sampling. Quality requires particles, purity at least 0.80 and fines fraction at most 0.50. The retained joint recommendation readout additionally applies a threshold of 0.10 to the implemented recovery output. The thermal direction comparison uses the original designated pair of recipes and retains its fixed scoring rule.

Recovery definition and protocol limitation

Tracing the executed observation and evaluator paths reveals a mismatch between the recovery description supplied to researchers and the response used for observation and scoring. The internal crystallization module stores a stage-recovery quantity: target recovered from solution, excluding seed, divided by target product present before separation. The active full-process observation path overrides that quantity with seed-excluded solid target divided by the tracked initial reagent charge, clipped to [0,1]. Source final assays report a noisy observation of this latter response. Recommendation retests and held-out references capture the same full-process response before final-assay sampling, and forecast evaluation reads those captured values. They do not read the internal stage-recovery value.

Saved source instructions and prediction prompts in both process blocks instead describe recovery relative to target product before separation. The two denominators are not interchangeable. The discrepancy is present in the recorded execution ver-

sions of both blocks. Supplementary Information distinguishes the implemented observation, the internal module quantity and the participant-facing definition; a difference between those interfaces must not be interpreted as a cohort effect.

All original predictions, references, scores and failure dispositions are retained. Recovery errors remain reproducible comparisons with the implemented target, but cannot be interpreted as an unconfounded test of understanding the quantity requested in the prompt. Joint success counts describe recipes under the implemented threshold rather than performance under an unambiguously communicated objective. We therefore report particle/purity/fines feasibility separately and do not use recovery accuracy as central evidence of scientific transfer. Purity and fines definitions are unaffected by this denominator discrepancy, although the research paths were generated under the mismatched instructions. A future qualification claiming aligned recovery understanding requires a consistent interface and a fresh affected experimental block, not relabelling or replacing these completed results.

Quantitative and qualitative analysis

Each campaign's mean absolute error averages its relevant question errors. Equilibrium macro error equally weights normalized pH, dissociation fraction and precipitation signal. The three most dilute queries are compared with the other nine, using the earlier public-concentration grouping retained in the prospective process note. That grouping was exploratory in the original broad study; the other nine are not asserted to be wholly in-distribution. The 1 mM source-coverage threshold is descriptive rather than a physical boundary or a success rule. Concentration is total nominal reagent added divided by total solvent added, not inferred dissolved concentration.

Process-cohort arm summaries first average valid sessions within a world and then weight the five worlds equally. Session values and observed denominators accompany aggregates. Worlds recur across information conditions and repeats; batches, queries, responses and reference observations are nested and correlated. We do not use them as independent researcher samples or report inferential tests based on that substitution. Displayed forecast intervals are the agents' original intervals, not confidence intervals for our aggregate means. Three sessions per cell describe variation without establishing a population success probability.

A single analyst reads all 284 source-note revisions, compares relevant sealed reports and checks the key case actions. The earlier 48-session reading and later 42-session reading are retained separately, including all ninety cases and version-specific quotations. Close reading distinguishes the world response, information available when a statement appears, the relationship and scope asserted, and its subsequent public use. Supplementary Information now includes an all-session inventory of notebook availability and assessment completion, followed by structured focal records separating prior expectation, executed test, revision, supported scope and subsequent use. Short quotations are checked against the specified note revision or assessment stage. These records document the selected relationships; the inventory is not a complete mechanistic coding of all ninety sessions.

Supported revisions, appropriate uncertainty, scope expansion and execution-report discrepancies are retained without an automated reasoning score. A missing or unexamined relation is marked as unresolved or not systematically assessed rather than absent. Intermediate-note gaps limit temporal resolution. No second independent coding or reliability estimate has been completed, and no prevalence of the qualitative mechanisms is inferred from the selected cases.

Main cases are retrospectively selected for an observable prospective commitment, relevant executed test and subsequent account, or for a clear link between a source relationship and later forecast. Counterexamples preserve successful extrapolation, correct treatment of rejected actions and appropriately bounded conclusions. Source action lists can include rejected attempts; successful seed additions are verified from transaction status. The record of forty 0.05 g requests therefore does not imply 2 g was added. Public rationale and temporal alignment support interpretation of an observed trajectory, not causal mediation by a statement or direct access to private cognition.

Matched quench intervention

The control block fixes all eighteen crystallization recommendations from the first process cohort before observing intervention outcomes. Each recipe is executed in all five original worlds with quench present and absent, giving ninety pairs and 180 trajectories. If the recipe includes quench, its original position defines the deletion; otherwise quench is inserted immediately before the first seed or crystallization-cooling operation. After removing that single operation, the two action lists are identical. Both arms start from the same world and initial state, use the same fixed observation identity, and preserve the recipe's materials, durations, measurements and other action arguments. No agent adapts after intervention.

All 180 trajectories are retained and replay-verified. Seventy-five pairs yield both final measurements. In five pairs the absent arm lacks particles for measurement; in ten the present arm makes the next fixed cooling target higher than the current temperature. Failed actions are not repaired and missing final outcomes are not imputed. Complete-pair effects therefore describe a restricted subset, alongside the full failure disposition. The primary contrast uses noiseless fines and recovery. Quench changes temperature, reaction-stop state, time and subsequent physical conditions; its total effect does not isolate cooling rate or nucleation. Repeated worlds and common donor recipes also preclude treating ninety pairs as ninety independent researchers.

The average present-minus-absent fines change is 0.30937 and recovery change 0.00741 in the seventy-five complete pairs. Joint quality/recovery success changes from 34 to 4. Fifteen of the eighteen original-world pairs are complete and are distinguished from transport to other worlds. The original-world pair for the selected first-cohort thermal-switch case fails after quench insertion; it has no final paired fines effect. New extension recipes are not donors to this control block. The intervention concerns environment outcomes, not the cause of an agent's scientific account.

Data availability

Source observations, versioned scientific records, sealed assessments and evaluator-derived summaries supporting this study are retained in the project evidence records. Supplementary Information provides all process-session outcomes, failure dispositions and retained broad-study tables. The integrated public archival package and its persistent identifier have not yet been finalized; the earlier public platform release should not be assumed to contain these later research cohorts.

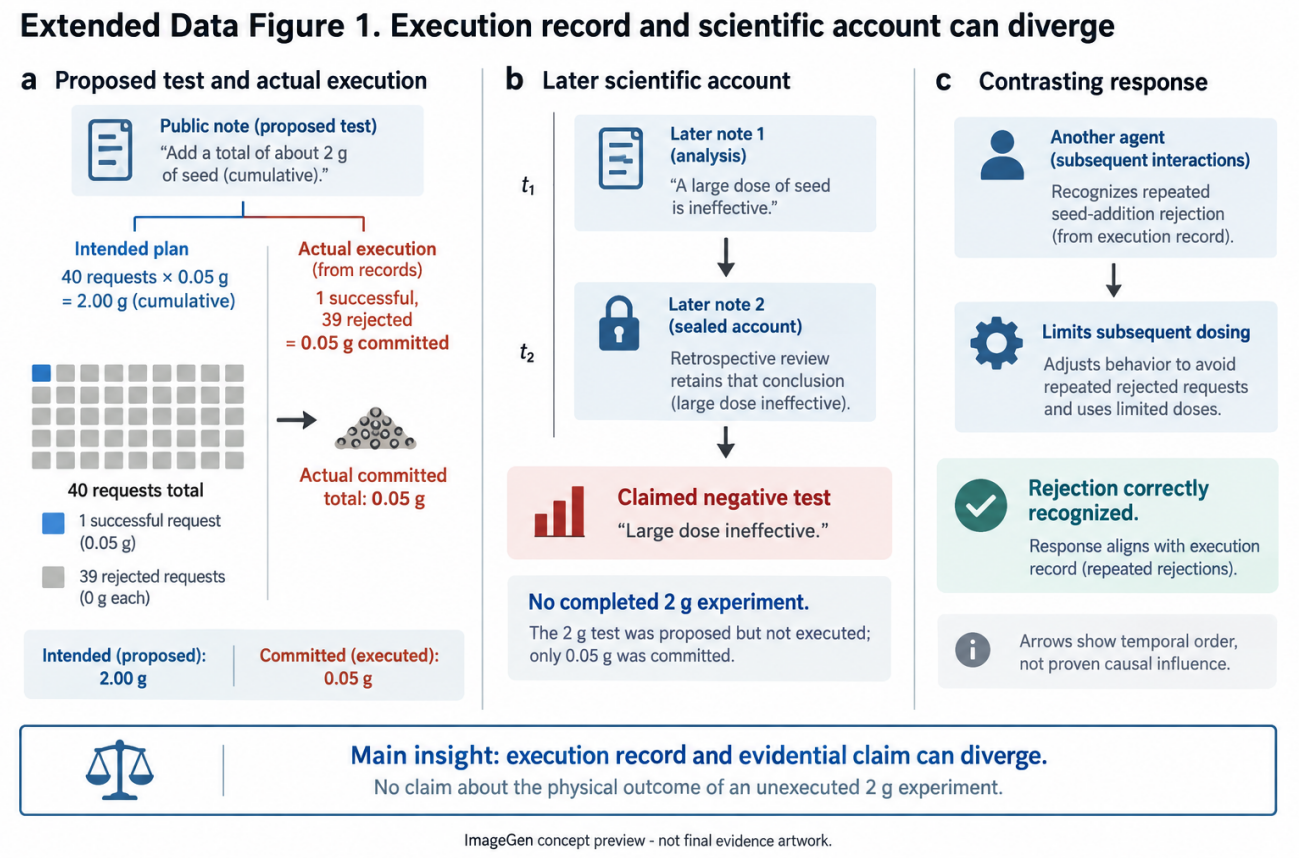
Code availability

Environment, execution, replay, scoring and analysis code are maintained in the ChemWorld project. The manuscript's analysis reads retained outputs without new participant or simulator calls. Exact execution versions and the source locations of every display are recorded in the accompanying evidence organization. A versioned public archive covering the integrated study remains to be finalized before submission.

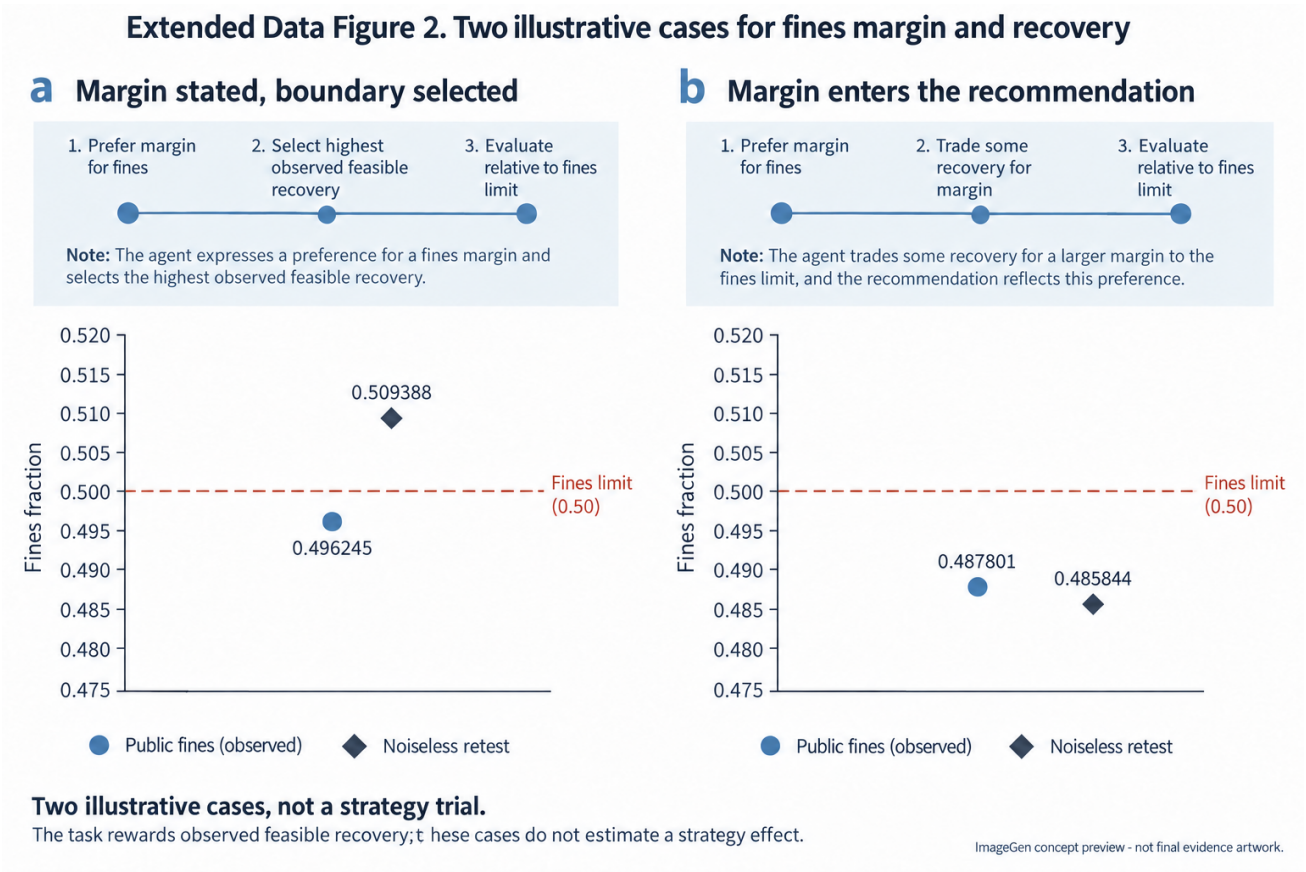
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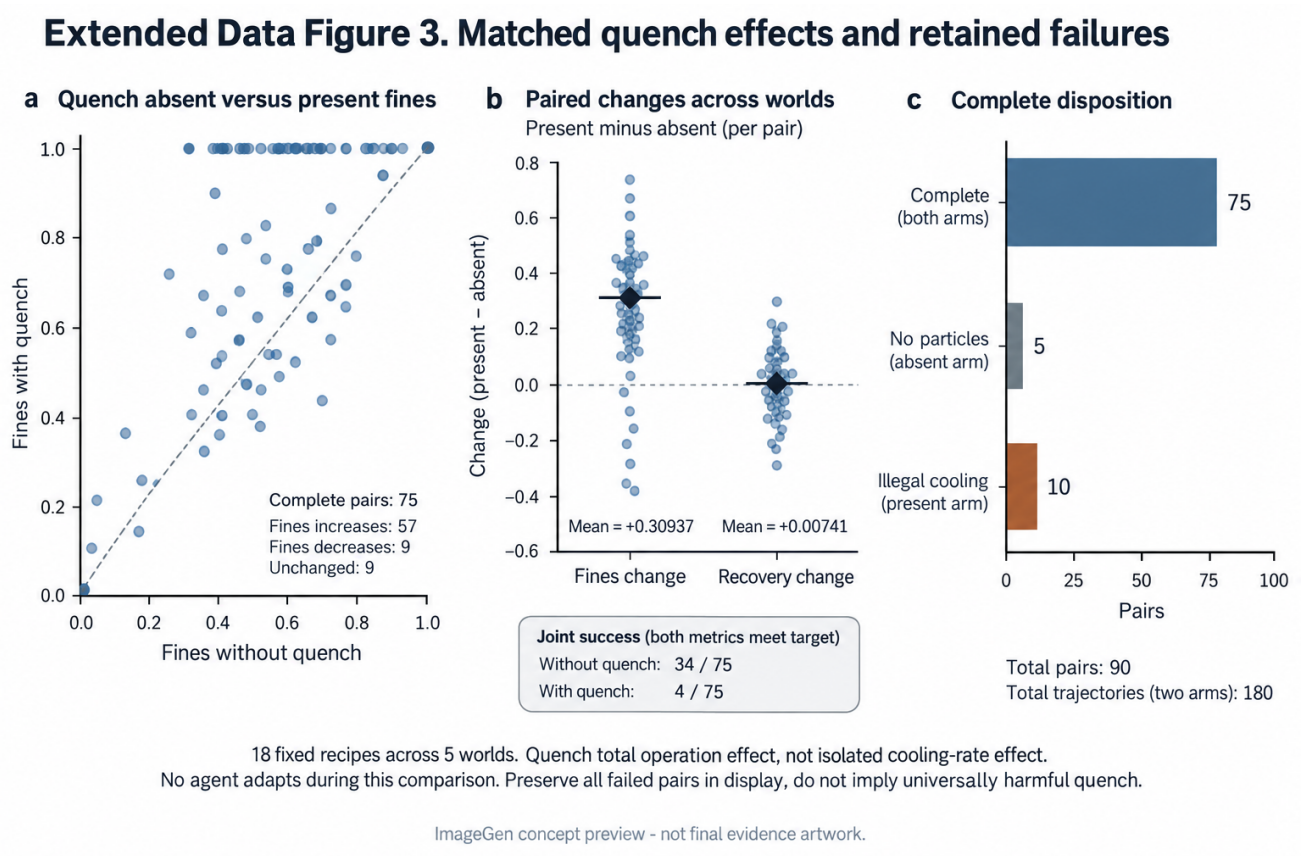
Extended Data



Extended Data Figure 1 | IMAGEGEN TEMPORARY PREVIEW. Generated marks and geometry are illustrative; retained tables and text carry the numerical evidence. An intended intervention becomes a stated negative test. a, A crystallization researcher proposes approximately 2 g of cumulative seed before the relevant batch. Forty additions of 0.05 g are requested, but only one commits successfully; thirty-nine are rejected. b, Subsequent notes and both sealed scientific accounts nevertheless retain the large-dose comparison as negative evidence. Text condenses public records; arrows indicate order rather than causal influence. c, A separate agent explicitly recognizes repeated seed-addition rejection and limits its subsequent claim. The first case does not establish the physical effect of 2 g because that dose was never executed.



Extended Data Figure 2 | IMAGEGEN TEMPORARY PREVIEW. Generated marks and geometry are illustrative; retained tables and text carry the numerical evidence. Stated uncertainty and operating choices. a, One researcher expresses a preference for fines margin, then recommends the highest observed feasible recovery near the quality boundary. Its public and noiseless retest fines fall on opposite sides of the limit. b, Another researcher explicitly trades some observed recovery for more margin and its recommendation remains feasible. Points distinguish public measurements and noiseless retests; the line denotes the fines limit. The two cases are illustrative rather than a randomized comparison of decision strategies. The task objective permits selecting the highest observed feasible recovery. Recovery values and rankings refer to the implemented initial-charge response under the disclosed prompt mismatch.



Extended Data Figure 3 | IMAGEGEN TEMPORARY PREVIEW. Generated marks and geometry are illustrative; retained tables and text carry the numerical evidence. Matched quench effects and incomplete pairs. a, Noiseless fines in quench-absent and quench-present arms for each of the 75 complete pairs; the diagonal denotes equality. b, Present-minus-absent mean fines and recovery effects and joint success counts; the generated distributions are illustrative, not measured point locations. c, All 90 planned pairs: complete measurement, absent-arm particle failure and present-arm cooling illegality. The control covers 18 first-cohort recommendations transported across five worlds; extension-cohort recipes are not included. The quench intervention changes the operation's total consequences, not temperature or reaction stop in isolation. Recovery changes and joint counts use the implemented initial-charge output. The prompt mismatch limits agent-performance interpretations, but does not change this fixed-recipe physical contrast.